MILITARY SPECIFICATION

CLOTH, COATED, CHLOROBUTYL, ARAMID, FUEL AND OXIDIZER RESISTANT

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers chlorobutyl coated aramid cloth
* for the fabrication of clothing for protection from propellants.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-P-1136	-	Packaging and Packing of Coated (Plastic, Rubber) and Laminated Fabrics.

MILITARY

* MIL-P-25604	-	Propellant, Unsymmetrical dimethylhydrazine.
MIL-P-26539	-	Propellant, Nitrogen Tetroxide.
MIL-P-27402	-	Propellant, Hydrazine - Unsymmetrical dimethyl-hydrazine (50 Percent N_2H_4 - 50 Percent UDMH).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFWAL/MLSE, ATTN: Standardization Manager, Wright-Patterson AFB, OH 45433-6533 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARD

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

NIOSH method s237 - The NIOSH Manual of Analytical Methods - Hydrazine.

(Application for copies should be addressed to the National Institute for Occupational Safety and Health, Robert A. Taft Laboratories, 4676 Columbia Parkway, Cincinnati OH 45226.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1385 - Test Method for Hydrazine in Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Preproduction sample approval. Prior to production, the contractor shall submit for approval the preproduction sample specified in 4.3.1.

3.2 Materials.

3.2.1 Yarn. The yarn shall be a nonmelting, aramid when tested as specified in 4.6.2.

3.2.2 Base fabric. The base fabric shall conform to the physical property requirements of table I when tested as specified in 4.6.1.

TABLE I. Base fabric.

Physical properties	Requirements
Thread count per inch (cm), minimum	
Warp	64 (25)
Filling	64 (25)
Thickness, inch (mm), maximum	0.009 (0.23)
* minimum	0.006 (0.15)
Weight per square yard, ounce (g per m^2)	3.5 ±0.5 (119 ±17)
Weave	2/2 basket
* Breaking strength, pound (N), minimum	
Warp	150 (667)
Filling	150 (667)
Charring point, minimum	Over 675°F (357°C)

* 3.2.3 Coating compound. The compound used for coating the base fabric shall be a composition of chlorobutyl rubber pigmented to produce an off-white to tan color on the heavy side and a blue color on the light side and formulated in accordance with established principles to meet the requirements of this specification.

* 3.2.4 Coating. The coating shall be unbalanced and applied to the base fabric at a nominal 2:1 ratio after which the coating shall be fully cured, for example: 325°F (163°C) for 2 hours.

3.3 Coated fabric. The coated fabric shall conform to the physical property and chemical permeation resistance requirements in table II when tested as specified in 4.6.3 and 4.7.

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TABLE II. Coated fabric.

Physical properties	Requirements
Thickness, inch (mm)	
Maximum	0.021 (0.53)
Minimum	0.017 (0.43)
Weight per square yard, ounce (g per m^2)	
Minimum	17.0 (576)
Maximum	23.0 (779)
Width (including selvage), inch (m)	
Minimum	39 (1)
Breaking strength, pound (N), minimum	
Initial (grab)	
Warp	150 (667)
Filling	150 (667)
Tearing strength, pound (N), minimum	
Warp	6 (27)
Filling	6 (27)
Abrasion resistance H-18 wheel, 500 g per wheel	
Minimum cycles with no fabric exposure	300
Blocking, 30 minutes at 180°F (82°C)	No blocking
Low temperature 4 hours at -40°F (-40°C)	No cracking, 180° bend on self
Flame resistance, 11 seconds exposure, 1½ inch (3.8 cm) flame	
After flame, seconds, maximum	15
Char length, inch (cm), maximum	4.5 (11.4)
Adhesion of coating, pound per inch (N per m), minimum	8.25 (1445)
Toxic gas permeability	
Oxidizer (MIL-P-26539) - The permeated concentration of NO_2 shall remain below the breakthrough concentration, defined by the equation in 4.7.4a, for more than 60 minutes when one side of the material	

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TABLE II. Coated fabric - Continued.

Physical properties	Requirements
Toxic gas permeability (continued)	
is exposed to 99 percent NO_2 vapor at the worst case permeation test cell temperature as determined by 4.7.4. Therefore, the time in minutes that the material must resist breakthrough is:	60
Fuel (MIL-P-27402) - Exposure at a saturated fume condition (maximum leakage), mg per inch² per hour (mg per cm² per hour)	
Hydrazine	0.005 (0.00077)
Unsymmetrical dimethylhydrazine (UDMH)	0.005 (0.00077)

3.4 Length. Each roll of coated fabric shall consist of not more than 3 pieces, the total length of which shall be 100 ±10 yards (91.44 ±9.14 meters [m]). No single piece shall be less than 10 yards (9.14 m).

3.5 Odor. The coated fabric shall be essentially odorless when examined as specified in 4.5.2.

3.6 Workmanship. The finished cloth shall be clean, evenly woven, and shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the cloth shall be classified as follows:

a. Preproduction test (see 4.3).

b. Acceptance tests (see 4.4).

4.3 Preproduction tests.

4.3.1 Preproduction test sample. The preproduction test sample (see 3.1) shall consist of a 5-yard (4.57 m) sample of the coated fabric representative of the production material.

* 4.3.2 Preproduction test report. With each preproduction sample the contractor shall furnish a certified test report to the Government, showing that the base and coated fabrics have met the requirements specified in 3.2 and 3.3, and have satisfactorily passed the preproduction tests specified in 4.3.3.

4.3.3 Preproduction tests. The preproduction tests shall consist of the overall examination for shade and finish (see 4.5.2) and all acceptance tests as described under 4.6 and 4.7.

4.4 Acceptance tests. Acceptance tests shall consist of the examinations specified in 4.5 and the acceptance tests specified in 4.6 and 4.7.

4.4.1 Lot size. For purposes of sampling, an inspection lot shall consist of all of the coated fabric produced on a continuous production run. Unscheduled production interruption shall constitute designation of another lot identification number.

4.5 Examination of the end item. Defects found during the examination shall be classified as specified herein.

4.5.1 Yard-by-yard examination. All yardage of each lot shall be inspected on both sides and visual defects classified as listed below. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The maximum acceptable defects shall be

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6 major defects and 12 total defects per roll length. All major and minor defects shall be permanently marked on the edge of the cloth opposite the * defect. All major defects shall also be permanently marked directly on the flaw.

Defect	Major	Minor
Any cut, hole, pinhole, tear or coated surface scratch or abrasion 1/	X	
Any uncoated area	X	
Any blister, tunnel, or delamination of coating	X	
Any lump or heavily coated area		X
Crease or wrinkle - resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure	X	
Uneven coating - thin area where coating is missing or insufficient	X	
Window - due to absence or poor blending of pigmentation only		X
Fabric edges rolled, folded, doubled or scalloped		X
Any permanent spot, stain, or streak or any spot, stain or streak from oil or grease resulting in defective coating	X	
Removable stain, spot or streak not affecting coating		X
Defects not marked or tagged	X	

1/ Examination for pinholes shall be made by viewing the surface of the fabric held under slight tension. A bright light source shall be positioned behind the coated fabric so as to clearly illuminate the surface without producing glare in the observer's eyes. A pinhole is defined as any opening observed in the film under the conditions specified above even though it may not be visible when the coated fabric is viewed normally in average daylight or the equivalent thereof.

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4.5.2 Overall examination. Each defect listed below shall be counted not more than once in each roll examined. The unit of product shall be one roll. The sample size (number of rolls) and the acceptance number shall be as shown in table III. The acceptance number is the maximum number of rolls that are found during the examination of the material to have a defect as listed below and the lot still be accepted by the Government. The rolls shall be examined on one side only, alternating the face and back sides.

Defects

Overall uncleanliness
Objectionable odor
Tackiness (coating will adhere and not readily unroll)

TABLE III. Sample size.

Lot size		Sample size (rolls)	Acceptance number
Yards	Meters		
1300 or less 1/	1188.72 or less	3	0
1301 to 3200	1189.63 to 2926.08	5	0
3201 to 8000	2926.99 to 7315.20	7	0
8001 to 22,000	7316.11 to 20 116.80	10	0
22,001 to 110,000	20 117.71 to 100 584	15	0
110,001 and up	100 584.91 and up	25	1

1/ If a lot contains fewer than 5 rolls, each roll in the lot shall be examined.

4.5.3 Examination for length. Each individual roll in the sample shall be examined for the defects listed below. The unit of product shall be 1 roll. The sample size and the acceptance number shall be as shown in table III.

Defects

Any roll (gross length) less than 90 yards (82.29 m) or more than 110 yards (100.84 m)
Any roll more than 2 yards (1.83 m) below gross length marked on ticket.
Any roll containing more than 3 pieces.
Any piece less than 10 yards (9.144 m).
Actual gross length of rolls less than the total gross yards marked on tickets.

4.5.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance

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with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units, as specified in MIL-STD-105.

Examination	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any nonconforming component, component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulged or distorted containers.
Weight	Gross weight exceeds requirements.

4.6 Physical acceptance tests.

4.6.1 Testing of base fabric. The methods used for testing of the base fabric shall be in accordance with table IV and 4.6.2. The sample unit shall be 1 continuous yard (0.9144 m). (Also see table I.)

TABLE IV. Tests of the base fabric.

Characteristic	Test method (FED-STD-191)	Determinations per unit	Results reported as:	
			Pass or fail	to the nearest
Thread count per inch				
Warp	5050	5		whole number
Filling	5050	5		
Thickness	5030	5		0.001 in. (0.025 mm)
Weight, ounce per square yard	5041	5		0.1 oz. (2.8 g)
Weave	visual	1	X	

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TABLE IV. Tests of the base fabric (Continued).

Characteristic	Test method (FED-STD-191)	Determinations per unit	Results reported as:	
			Pass or fail	to the nearest
Breaking strength				
Warp	5100	5		1.0 lb. (0.45 kg)
Filling	5100	5		1.0 lb. (0.45 kg)
Charring	(see 4.6.2)	1	X	

4.6.2 Nonmelting of yarn.

4.6.2.1 Apparatus. The apparatus shall consist essentially of an electrically heated stage having a circular depression, a powerstat controlling the rate of heat input into the stage, and a thermometer inserted in the stage (see 6.4).

4.6.2.2 Procedure. The specimen, one pick and one end from the base fabric, shall be placed in the circular depression on the stage and cover glass placed over the specimen. The temperature of the stage shall be raised at a rapid rate to 662°F (350°C) and thereafter at a rate of 3 to 4 degrees per minute up to 675°F (357°C).

4.6.3 Testing of the end item. The methods used for testing the coated fabric shall be in accordance with table V, and 4.7. The requirements specified in table II apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. For those characteristics wherein test results are reported as pass or fail, there shall be no evidence of failure of any single determination. The sample unit shall be 5 continuous yards (4.57 m) full width of the finished fabric and only 1 sample shall be taken from each roll. The lot shall be unacceptable if one or more units fail to meet any requirement of the specification. The number of samples shall be as follows:

(Yards)	(Meters)	Number of samples
500 or less	457.22 or less	1
501 to 1000	458.11 to 914.40	2
1001 and over	915.31 and over	3

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TABLE V. Tests of the coated fabric.

Characteristic	Test method (FED-STD-191)	Determinations per unit	Results reported as:	
			Pass or fail	to the nearest
Thickness	5030	5		0.001 in. (0.025 mm)
Weight, ounce per square yard	5041	5		0.1 oz. (2.85 g)
Width	5020	3	X	
Breaking strength (each direction)				
Initial	5100	5		1.0 lb. (0.45 kg)
After weathering	5100 & 1/ 5804	5		1.0 lb. (0.45 kg)
Tearing strength	5134	5		0.1 lb. (45 g)
Abrasion resistance	5306	4	X	
Blocking	5872	3	X	
Low temperature effect	2/ 5874	3	X	
Flame resistance	3/ 5904			
After flame		5		0.2 seconds
Char length		5		0.1 in. (2.5 mm)
Adhesion of coating	4/ 5970	4		0.1 lb. (45 g)
Toxic gas permeability				
Oxidizer	(see 4.7.4)	3		0.00015 mg/cm²
Fuel (hydrazine)	(see 4.7.5)	2		0.00015 mg/cm²
Fuel (UDMH)	(see 4.7.6)	2		0.00015 mg/cm²

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1/ Exposure time shall be 150 hours.

2/ Except that the specimen shall be exposed at a temperature of -40°F (-40°C) for a period of 4 hours.

3/ The test shall be conducted in an inclosed cabinet such as that described in method 5902.

4/ If the coating is separated from the base cloth at an average force greater than that specified or if the average force necessary to separate the cemented sample at the cement line is greater than that specified, the test for adhesion of coating shall be considered passed.

4.7 Chemical permeation resistance.

4.7.1 Fuel. The fuel used in 4.7.3, 4.7.4, 4.7.5 and 4.7.6 shall be in accordance with MIL-P-27402.

4.7.2 Oxidizer. The oxidizer used in 4.7.3, 4.7.4, 4.7.5 and 4.7.6 shall be in accordance with MIL-P-26539.

* 4.7.3 Preparation of test specimens. In order to minimize the potential for contamination, the test specimens shall be prepared no more than 3 days ahead of the start of their test sequence.

a. Visually examine the material to ensure that there are no observable flaws.

b. Measure and record the thickness of the material.

c. Avoid contamination of the sample by using tweezers in all manipulations and by avoiding any contact with countertops.

d. Rinse the material with deionized water to remove any particulates or foreign materials.

e. Hang the material in a clean location for at least 16 hours and no more than 3 days to allow it to air-dry.

f. Place the material on a clean sheet of polyethylene and cut a 10 cm by 10 cm (4 inch by 4 inch) square of material (see figure 1) immediately prior to the start of its evaluation.

* 4.7.3.1 Conditioning. The conditioning block that is used in the following procedures is detailed in figure 2.

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a. Oxidizer exposure.

(1) The specimen shall be installed in the clean, dry, leveled (Theta [angle] equals 180°) conditioning block (see figures 2 and 3) with its intended outside facing upwards towards the Teflon block and the intended inside laying within the machined indent on the stainless steel base plate (see figure 2). Assemble the conditioning block and secure the material in place by tightening the four bolts that are located at the corners of the conditioning block (see figure 2). Adjust the temperature of both the hypergol reservoir and the conditioning block to 15°C. In order to prevent premature exposure of the material to the oxidizer vapors, verify that solenoid valve SV-1 and manual valve MV-5 are closed.

(2) Turn three-way valve TW-7 towards the N_2O_4 cylinder and carefully open manual valve MV-8 to allow 50 mL of liquid oxidizer to condense in the chilled hypergol reservoir. Before proceeding further, verify that the valves are in the following positions: SV-1 closed, SV-2 closed, MV-3 closed, SV-4 closed, MV-5 closed, SV-6 closed, TW-7 towards N_2O_4 cylinder, and MV-8 closed. Verify that the water pump is primed and that the water reservoir is filled.

(3) Initiate a 1-minute immersion of the intended outside of the material in 50 mL of chilled liquid oxidizer by opening manual valve MV-5 and then solenoid valve SV-1. Allow the liquid propellant to completely drain onto the material and then turn on the water pump (the water will not flow until SV-6 is opened) and shut SV-1. After 60 seconds of exposure to the liquid propellant, lower (Theta equals 115°) the conditioning block, and initiate 1 minute of water rinsing by opening solenoid valves SV-2 and SV-6. After 1 minute of water rinsing (at 1 liter per minute), implement an air purge of the system by shutting SV-6, opening SV-4, and adjusting MV-3 to obtain an air flow of 5 liters per minute. Turn off the water pump.

(4) After the system has been purged for 5 minutes with dry air, shut SV-4, remove the specimen from the conditioning block, and dunk it successively into 3 beakers of deionized water for 15, 15, and 30 seconds, respectively. Hang the rinsed material in a clean, dry place and allow to dry for at least 16 hours and no more than 3 days. Repeat this sequence for a total of 6 exposures for each specimen.

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b. Fuel exposure.

(1) The specimen shall be installed in the clean, dry, leveled (Theta equals 180°) conditioning block (see figures 2 and 4) with its intended outside facing upwards towards the Teflon block and the intended inside laying within the machined indent on the stainless steel base plate (see figure 2). Assemble the conditioning block and secure the material in place by tightening the 4 bolts that are located at the corners of the conditioning block (see figure 2). Adjust the temperature of both the hypergol reservoir and the conditioning block to 15°C. In order to prevent premature exposure of the material to the fuel vapors, verify that manual valves MV-1 and MV-5 are closed.

(2) Transfer 50 mL of liquid fuel into the chilled hypergol reservoir. Before proceeding further, verify that the valves are in the following positions: MV-1 closed, MV-2 closed, MV-3 closed, SV-4 closed, MV-5 closed, and SV-6 closed. Verify that the water pump is primed and that the water reservoir is filled.

(3) Initiate a 1-minute immersion of the intended outside of the material in 50 mL of chilled liquid fuel by opening manual valve MV-5 and then manual valve MV-1. Allow the liquid propellant to completely drain onto the material and then on the water pump (the water will not flow until SV-6 is opened) and shut MV-1. After 60 seconds of exposure to the liquid propellant, lower (Theta equals 115°) the conditioning block, and initiate 1 minute of water rinsing by opening manual valve MV-2 and solenoid valve SV-6. After 1 minute of water rinsing (at 1 liter per minute), implement an air purge of the system by shutting solenoid valve SV-6, opening solenoid valve SV-4, and adjusting manual valve MV-3 to obtain an air flow of 5 liters per minute. Turn off the water pump.

(4) After the system has been purged for 5 minutes with dry air, shut SV-4, remove the specimen from the conditioning block, and dunk it successively into 3 beakers of deionized water for 15, 15 and 30 seconds, respectively. Hang the rinsed material in a clean, dry place and allow it to dry for at least 20 hours and no more than 3 days. Repeat this sequence for a total of 6 exposures for each specimen.

c. Alternating exposures. The specimen shall be subjected to 1 exposure according to the procedure described in 4.7.3.1a followed by 1 exposure according to the procedure described in 4.7.3.1b. Repeat the alternating exposures for a total of 6 exposures (i.e., a

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total of 3 oxidizer exposures and a total of 3 fuel exposures) for each specimen.

WARNING: The fuel and oxidizer are highly toxic; handle with care.

* 4.7.4 Oxidizer permeability test. In order to determine the worst case temperature, two unconditioned specimens shall be evaluated by the following test procedure at two different permeation test cell temperatures: one sample at 22°C and another sample at 35°C. The NO_2 breakthrough times (as defined in 4.7.4a) for these two experiments shall be used to establish the worst case permeation test cell temperature to be used in the remaining oxidizer permeation tests. Three specimens each conditioned in accordance with 4.7.3.1a, b, and c and three unconditioned specimens shall then be evaluated by the following test procedure at the worst case permeation cell temperature (i.e., the permeation test cell temperature that resulted in the shortest breakthrough time in the above tests).

a. Real-time NO_2 detectors shall be used to monitor the oxidizer concentration in the sweep air (i.e., the air sweeping across the unexposed side of the material) until either NO_2 breakthrough or until 6 hours have elapsed. Breakthrough is defined as attainment of a critical concentration of propellant in the sweep air. The critical breakthrough concentration is defined as the concentration of propellant calculated from the following equation:

$$C_B = \frac{16.667 \times A \times R \times T}{F \times P \times Mw}$$

where: X is the maximum cumulative permeation (0.00154 mg/cm² hour for NO_2) during the first 60 minutes of testing

A is the area (25 cm²) of the material that is exposed to the propellant

R is the gas constant (0.08205 1 atm/K mole)

T is the absolute temperature (Kelvin)

F is the flow rate of sweep air (2.0 liters per minute) across the unexposed side of the material

P is the pressure (1.0132 atmospheres) in the permeation test cell

Mw is the gram-molecular weight of the propellant (46.0 for NO_2).

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As apparent from the above equation, the critical concentration depends upon several of the experimental parameters (T, P, and F), on the identity of the propellant (Mw), and upon the area (A) of the material. For the values of the parameters given in parentheses, the critical concentration and hence the detector's lower detection limit will be 0.167 ppm for 295.16 K (22°C) and 0.174 ppm for 308.16 K (35°C). Therefore, 0.17 ppm will be the breakthrough concentration for both of these experiments. In order to be certain that the breakthrough is not a temporary instability in the detector's response, the NO_2 concentration in the sweep air shall be monitored until it exceeds the upper range of the detector (which shall be at least 10 times the breakthrough concentration).

b. Assemble the clean, dry permeation test apparatus as illustrated in figure 5. Install the test material between the Pyrex joints (2 modified glass joints which have stems that are 50 mm inside diameter and 54 mm outside diameter) of the permeation test cell with the intended outside of the material facing downward and the intended inside of the material facing upward. Use of butyl rubber o-ring (size 2-229) to form an air-tight seal on the upper surface (intended inside) of the material as indicated in figure 5. Before proceeding further, verify that the valves are in the following positions: MV-1 open, MV-2 shut, MV-3 shut, MV-4 shut, MV-5 open, MV-6 shut, MV-7 open, MV-8 shut, SV-9 shut, SV-10 shut, SV-11 shut, TW-12 towards vent, MV-13 shut, TW-14 towards vent, SV-15 shut, and MV-16 shut. Adjust the zero on the differential capacitance manometer. Turn on the vacuum pump and open MV-4. Wait for the system to pump down to ~0.1 mbar and adjust the zero on the absolute manometer to obtain the agreement between the thermocouple vacuum gauge and the absolute capacitance manometer. Shut MV-4, open MV-3, open MV-6, and shut MV-7. Record the absolute pressure which is the ambient atmospheric pressure.

c. Adjust the sweep air flow to 2.0 liters per minute using MV-2. Open SV-9 and adjust the air flow through the bottom of the test cell to 70 mL per minute using MV-8. Using MV-1 and MV-5 to restrict the air flow from the upper and lower half of the test cell, respectively, simultaneously adjust the absolute pressure 102,6000 Pa (770 torr) while maintaining a differential pressure of 75 Pa (0.56 torr or 0.3 inches of water) across the material sample. A positive differential pressure is defined as a higher pressure on the intended inside (i.e., the upper half of the test cell as drawn in figure 5) than on the intended outside of the material sample.

d. Turn on the fan and adjust the temperatures of the 2 circulating constant temperature baths to obtain the test cell temperature that was specified in 4.7.4 and 21.3°C for the N_2O_4 bubbler (which

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corresponds to 99 percent oxidizer in the 102,600 Pa total pressure). Introduce liquid N_2O_4 into the N_2O_4 bubbler by opening SV-11, turning TW-12 towards the N_2O_4 cylinder, and carefully opening MV-13 until liquid oxidizer starts to flow from the 25°C N_2O_4 cylinder into the bubbler. After collecting approximately 100 mL of liquid oxidizer, shut MV-13. Purge the liquid oxidizer with air for 30 minutes by opening SV-15, adjusting the air flow to 0.5 mL per minute with MV-16, and diverting the air through the bubbler by turning TW-14 towards the bubbler.

e. Calibrate the NO_2 detector by challenging it with span gas containing a verified concentration of NO_2 vapor. The span gas concentration can be at any concentration within the calibrated range of the detector as long as the concentration has been verified by the method indicated in 4.7.4.1. Establish the baseline for the experiment by allowing the detector to sample the sweep air from the sampling port (see figure 5) and recording its response on a strip chart recorder (chart speed shall be at least 6 cm per hour).

f. Maintain the specified temperatures (±0.5°C), pressures (±100 Pa for the absolute pressure and ±10 Pa for the differential pressure) and air flows (±2 percent) throughout the experiment.

g. Initiate the exposure of the intended outside of the material to the 99 percent NO_2 vapor by simultaneously opening SV-10 while shutting SV-9 and SV-11, marking the start of exposure on the strip chart recording. If needed, adjust MV-5 to maintain 75 Pa of differential pressure across the material. Close MV-8. Maintain the experimental parameters at their specified values while monitoring the concentration on NO_2 in the sweep air. Record the time elapsed between the initiation of exposure to oxidizer and the attainment of the breakthrough concentration of 0.17 ppm in the sweep air. Continue to monitor the NO_2 concentration until it exceeds the range of the detector.

h. Allow the detector to recover while sampling clean, dry air. Stop the flow of oxidizer to the permeation test cell by simultaneously shutting SV-10 while opening SV-9 and SV-11. Adjust MV-8 to obtain a flow of 70 mL per minute of air through the bottom of the permeation test cell. Turn TW-14 towards the vent, shut MV-16 and SV-15. Simultaneously open MV-1 and MV-5 to reduce the pressure in the permeation test cell. Pump out the capacitance manometers by shutting MV-3, shutting MV-6, opening MV-7, and opening MV-4 (with the vacuum pump remaining on throughout the experiment). After the oxidizer vapors have been purged from the permeation test cell, shut

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MV-2, MV-8, and SV-9. Remove the test material from the permeation test cell and rinse it in deionized water. Allow it to hang in an uncontaminated section of the hood to air-dry overnight.

i. Dispose of the excess oxidizer, the oxidizer contaminated waste water, and all oxidizer contaminated tubing. Clean the permeation test cell, the waste reservoir, and the bubbler. Repeat the test for all the material samples that are specified in 4.7.4.

* 4.7.4.1 Verification of the concentration of the NO_2 calibration gas. The dinitrogen tetroxide calibration gas is drawn through a gas-absorbing bottle containing an absorbing reagent. The NO_2 which has been absorbed is determined by measuring the optical density (percent transmission) using a colorimeter-spectrophotometer. The reaction follows Beer's Law in that the concentration is directly proportional to the optical density. The correlation between optical density and concentration is obtained by measuring the optical density of sodium nitrite solutions of varying known concentrations. A curve of optical density versus concentration of nitrogen tetroxide may then be prepared by using the established relationship that one mole of NO_2 yields the same color intensity as 0.72 moles of sodium nitrite. Once this curve has been prepared, the test procedure involves merely purging permeated vapors into absorbing solution, placing a sample of the colored solution in the colorimeter-spectrophotometer, and measuring the optical density. The quantity of nitrogen tetroxide which has permeated may then be read from the reference graph.

4.7.4.1.1 Apparatus.

a. Colorimeter-spectrophotometer, Bausch & Lomb Spectronic 20, or equal.

b. Fritted disc gas washing bottle. Note: The fritted disk washing bottle shall have pore frits with diameters no greater than 60 microns. Pore frit size affects the accuracy of this analytical method.

4.7.4.1.2 Reagents. All reagents are prepared from analytical grade chemicals in nitrite-free distilled water.

N-(1-naphthyl)-ethylenediamine dihydrochloride 0.1 percent stock solution. Prepare by dissolving 0.5 gram (g) in 500 mL of water.

4.7.4.1.2.1 Absorbing reagent. Place 140 mL of glacial acetic acid in a 1-liter volumetric flask containing approximately 750 mL of water. Dissolve 5 g of sulfanilic acid in this solution. Add 20 mL of the 0.1 stock solution of N-(1-naphthyl)-ethylenediamine dihydrochloride. Dilute to 1 liter.

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4.7.4.1.2.2 Standard sodium nitrite solution. Prepare a stock solution by dissolving approximately 7 g in 1 liter water. Using this stock solution, prepare the standard solution by diluting to approximately 0.035 g per liter. The standard solution shall be prepared fresh each time it is used.

4.7.4.1.3 Procedure.

a. Prepare the reagent solutions as indicated above.

b. Using the standard sodium nitrite solution, prepare reference samples of sodium nitrite in absorbing solution by placing samples of standard sodium nitrite solution into volumetric flasks and diluting with absorbing reagent. The range of concentration shall be approximately 1.2 (10^{-7}) g per mL to 9 (10^{-7}) g per mL.

c. Determine the optical density of each reference sample with the spectrophotometer set at a wave length of 550 nanometers (nm) and tabulate optical densities and corresponding sodium nitrite concentrations.

d. Convert the sodium nitrite concentration values to nitrogen tetroxide concentration by multiplying the $NaNO_2$ concentration (g per mL) by 0.926.

e. Plot nitrogen tetroxide concentration versus optical density.

f. From this graph select the concentration which corresponds to 0.30 optical density (approximate center scale).

g. To produce the center-scale reading at the maximum allowable permeation rate, calculate the quantity of absorbing reagent to be used in the permeability test using the following equation:

$$V = \frac{MA}{C}$$

where: V is the volume of absorbing reagent into which permeated nitrogen tetroxide vapors are passed (mL)

M is the maximum allowable permeation rate $\frac{(g)}{inch^2 - hour}$

A is the exposed area of component under test

C is the concentration of nitrogen tetroxide (g per m) which corresponds to an optical density of 0.3

h. Place V mL of absorbing reagent, as determined in step g above, into the gas washing bottle.

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i. Connect the inlet tube of the gas washing bottle to the purge line from the test component.

j. Purge continuously for the entire test duration.

k. The nitrogen tetroxide which has permeated the component is absorbed by the reagent. After the test is completed, allow 15 minutes for full color to develop.

l. Remove a sample of the exposed absorbing reagent and determine the optical density with the colorimeter-spectrophotometer (with wave length set at 550 nm).

m. Obtain the value of permeation to be reported from the graph prepared in step e in the following manner:

(1) Find the optical density on the graph from step e.

(2) Record the corresponding concentration in g per mL.

(3) Multiply the concentration by the total volume in mL of absorbing agent.

(4) Divide the area of fabric exposed to N_2O_4.

* 4.7.4.2 Real-time detectors for monitoring the concentration of dinitrogen tetroxide. Electrochemical and gas phase chemiluminescent detectors are applied to the real-time detection of permeated NO_2. The response of the detectors shall be calibrated by challenging them with span gases of several known concentrations, verified using the procedure in 4.7.4.1. Once the calibration curve (the detector response versus challenge gas concentration) are established for the detector, the detector shall be calibrated by challenging it with one span gas concentration that is within the calibrated range of the detector.

* 4.7.5 Fuel permeability test - hydrazine detection. In order to determine the worst case temperature, three unconditioned specimens shall be evaluated by the following test procedure at three different permeation test cell temperatures: the first sample at 15°C, the second at 22°C, and the third sample at 35°C. The hydrazine breakthrough times (as defined in 4.7.5a) for these three experiments shall be used to establish the worst case permeation test cell temperature to be used in the remaining hydrazine permeation tests. Two specimens each conditioned in accordance with 4.7.3.1a, b, and c and two unconditioned specimens shall then be evaluated by the following test procedure at the worst case permeation cell temperature (the permeation test cell temperature that resulted in the shortest breakthrough time in the above tests). In the latter tests the quantitative analysis of hydrazine permeation during the first hour of exposure shall be conducted in accordance with either ASTM D 1385 or NIOSH method s237.

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a. The maximum cumulative leakage of hydrazine vapor through the test material during the first hour of any hydrazine permeation test is listed in table II. In order to determine the worst case temperature for the permeation test, real-time hydrazine detectors shall be used to monitor the hydrazine concentration in the sweep air (the air sweeping across the unexposed side of the material). The hydrazine concentration in the sweep air shall be monitored until either hydrazine breakthrough or until 6 hours have elapsed. Breakthrough is defined as attainment of a critical concentration of propellant in the sweep air. In order to ensure that the amount of permeated hydrazine does not exceed the maximum leakage, the critical breakthrough concentration is defined as the concentration of propellant calculated from the equation in 4.7.4a using a value of 0.00077 mg per cm^2 hour (as specified in table II for hydrazine) for the maximum cumulative permeation (X in stated equation). For the values of the parameters given in 4.7.4a, the critical concentration and hence the detector's lower detection limit will be 0.117 ppm for 288.16K (15°C) and 0.1249 ppm for 308.16 K (35°C). Therefore, 0.12 ppm will be the breakthrough concentration for both of these experiments. In order to be certain that the breakthrough is not a temporary instability in the detector's response, the hydrazine concentration in the sweep air shall be monitored until it exceeds the upper range of the detector (which shall be at least 10 times the breakthrough concentration).

b. Assemble the clean, dry permeation test apparatus as illustrated in figure 6. Install the test material between the paraffin coated Pyrex joints (2 modified glass joints which have stems that are 50 mm inside diameter and 54 mm outside diameter) of the permeation test cell with the intended outside of the material facing upward and the intended inside of the material facing downward. Use 1 butyl rubber o-ring (size 2-229) to form an air-tight seal on the lower surface (intended inside) of the material as indicated in figure 6. Before proceeding further, verify that the valves are in the following positions: MV-1 closed, MV-2 closed, MV-3 closed, MV-4 open, MV-5 closed, MV-6 closed, and MV-7 open. Adjust the zero on the differential capacitance manometer. Turn on the vacuum pump and open MV-3. Wait for the system to pump down to ~0.1 mbar and adjust the zero on the absolute manometer to obtain the agreement between the thermocouple vacuum gauge and the absolute capacitance manometer. Shut MV-3, open MV-5, open MV-2, and shut MV-7. Record the absolute pressure (ambient atmospheric pressure).

c. Adjust the sweep air flow to 2.0 liters per minute using MV-6. Using MV-4 to restrict the air flow from the lower half of the test cell, adjust the differential pressure to 137 Pa (1.03 torr or 0.55 inches of water) across the sample of test material. Since the fuel will cover the material to a depth of 0.64 cm (0.25 inches), the

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actual differential pressure across the material will be 75 Pa (0.56 torr or 0.3 inches of water). A positive differential pressure is defined as a higher pressure on the intended inside (the lower half of the test cell as drawn in figure 6) than on the intended outside of the material sample.

d. Turn on the fan and adjust the temperature of the 2 circulating constant temperature baths to obtain the test cell temperature that was specified in 4.7.5; the hypergol reservoir shall be at the same temperature as the permeation test cell. Pour 16 mL of liquid fuel into the hypergol reservoir.

e. For the initial experiments that use a real-time detector, calibrate the hydrazine detector by challenging it with span gas containing a verified concentration of hydrazine vapor. The span gas concentration shall be no more than 2 times the breakthrough concentration and shall be verified by either ASTM D 1385 or NIOSH method s237. Establish the baseline for the experiment by allowing the detector to sample the sweep air from the sampling port (see figure 6) and recording its response on a strip chart recorder. The chart speed shall be at least 6 cm per hour.

f. Maintain the specified temperatures (±0.5°C), differential pressure (±10 Pa), and the sweep air flow (±2 percent) throughout the experiment.

g. Initiate the exposure of the intended outside of the material to the liquid fuel by opening MV-1, marking the start of exposure on the strip chart recording. If needed, adjust MV-4 to maintain the nominal 137 Pa of differential pressure across the material. Maintain the experimental parameters at their specified values while monitoring the concentration of hydrazine in the sweep air. Record the time elapsed between the initiation of exposure to hydrazine and the attainment of the breakthrough concentration of 0.12 ppm in the sweep air. Continue to monitor the hydrazine concentration until it exceeds the range of the detector.

h. Allow the detector to recover while sampling clean, dry air. Open MV-4 to reduce the differential pressure across the fabric. Pump out the capacitance manometers by shutting MV-2, shutting MV-5, opening MV-7, and opening MV-3 (with the vacuum pump remaining on throughout the experiment). Shut MV-6 and dispose of the fuel. Rinse the permeation test cell with deionized water and remove the test material from the permeation test cell. Rinse the material with deionized water. Allow it to hang in an uncontaminated section of the hood to air-dry overnight.

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i. Dispose of the hydrazine contaminated tubing. Clean the permeation test cell and the hypergol reservoir. Repeat the test for all of the material samples that are specified in 4.7.5.

* 4.7.6 Fuel permeability test - UDMH detection. In order to determine the worst case temperature, three unconditioned specimens shall be evaluated by the following test procedure at three different permeation test cell temperatures: the first sample at 15°C, the second at 22°C, and the third sample at 35°C. The UDMH breakthrough times (as defined in 4.7.6a) for these three experiments shall be used to establish the worst case permeation test cell temperature to be used in the remaining UDMH permeation tests. Two specimens each conditioned in accordance with 4.7.3.1a, b, and c and two unconditioned specimens shall then be evaluated by the following test procedure at the worst case permeation cell temperature (the permeation test cell temperature that resulted in the shortest breakthrough time in the above tests). In the latter tests the quantitative analysis of the UDMH permeation shall be carried out in accordance with either 4.7.7 or 4.7.8.

a. The maximum cumulative leakage of UDMH vapor through the test material during the first hour of the UDMH permeation test is listed in table II. In order to determine the worst case temperature for the permeation test, real-time UDMH detectors shall be used to monitor the UDMH concentration in the sweep air (the air sweeping across the unexposed side of the material). The UDMH concentration in the sweep air shall be monitored until either UDMH breakthrough or until 6 hours have elapsed. Breakthrough is defined as attainment of a critical concentration of propellant in the sweep air. In order to ensure that the amount of permeated UDMH does not exceed the maximum leakage, the critical breakthrough concentration is defined as the concentration of propellant calculated from the equation in 4.7.4a using a value of 0.00077 mg per cm^2 hour (as specified in table II for UDMH) for the maximum cumulative permeation (X in stated equation). For the values of the parameters given in 4.7.4a, the critical concentration and hence the detector's lower detection limit will be 0.062 ppm for 288.16 K (15°C), 0.064 ppm for 295.16 K (22°C) and 0.067 ppm for 308.16 K (35°C). Therefore, 0.06 ppm will be the breakthrough concentration for all three of these experiments. In order to be certain that the breakthrough is not a temporary instability in the detector's response, the UDMH concentration in the sweep air shall be monitored until it exceeds the upper range of the detector (which shall be at least 10 times the breakthrough concentration).

b. Assemble the clean, dry permeation test apparatus as illustrated in figure 6. Install the test material between the paraffin coated Pyrex joints (2 modified glass joints which have stems that are 50 mm inside diameter and 54 mm outside diameter) of the permeation test cell with the intended outside of the material facing upward

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and the intended inside of the material facing downward. Use one butyl rubber o-ring (size 2-229) to form an air-tight seal on the lower surface (intended inside) of the material as indicated in figure 6. Before proceeding further, verify that the valves are in the following positions: MV-1 closed, MV-2 closed, MV-3 closed, MV-4 open, MV-5 closed, MV-6 closed, and MV-7 open. Adjust the zero on the differential capacitance manometer. Turn on the vacuum pump and open MV-3. Wait for the system to pump down to ~0.1 mbar and adjust the zero on the absolute manometer to obtain the agreement between the thermocouple vacuum gauge and the absolute capacitance manometer. Shut MV-3, open MV-5, open MV-2, and shut MV-7. Record the absolute pressure (ambient atmospheric pressure).

c. Adjust the sweep air flow to 2.0 liters per minute using MV-6. Using MV-4 to restrict the air flow from the lower half of the test cell, adjust the differential pressure to 137 Pa (1.03 torr or 0.55 inches of water) across the sample of test material. Since the fuel will cover the material to a depth of 0.64 cm (0.25 inches), the actual differential pressure across the material will be 75 Pa (0.56 torr or 0.3 inches of water). A positive differential pressure is defined as a higher pressure on the intended inside (the lower half of the test cell as drawn in figure 6) than on the intended outside of the material sample.

d. Turn on the fan and adjust the temperatures of the 2 circulating constant temperature baths to obtain the test cell temperature that was specified in 4.7.6; the hypergol reservoir shall be at the same temperature as the permeation test cell. Pour 16 mL of liquid fuel into the hypergol reservoir.

e. For the initial experiments that use a real-time detector, calibrate the UDMH detector by challenging it with span gas containing a verified concentration of UDMH vapor. The span gas concentration shall be no more than 2 times the breakthrough concentration and shall be verified in accordance with 4.7.7 or 4.7.8. Establish the baseline for the experiment by allowing the detector to sample the sweep air from the sampling port (see figure 6) and recording its response on a strip chart recorder. The chart speed shall be at least 6 cm per hour.

f. Maintain the specified temperatures (±0.5°C), differential pressure (±10 Pa), and the sweep air flow (±2 percent) throughout the experiment.

g. Initiate the exposure of the intended outside of the material to the liquid fuel by opening MV-1, marking the start of exposure on the strip chart recording. If needed, adjust MV-4 to maintain the nominal 137 Pa of differential pressure across the material. Maintain the experimental parameters at their specified values while

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monitoring the concentration of UDMH in the sweep air. Record the time elapsed between the initiation of exposure to fuel and the attainment of the breakthrough concentration of 0.06 ppm in the sweep air. Continue to monitor the UDMH concentration until it exceeds the range of the detector.

h. Allow the detector to recover while sampling clean, dry air. Open MV-4 to reduce the differential pressure across the fabric. Pump out the capacitance manometers by shutting MV-2, shutting MV-5, opening MV-7, and opening MV-3 (with the vacuum pump remaining on throughout the experiment). Shut MV-6 and dispose of the fuel. Rinse the permeation test cell with deionized water and remove the test material from the permeation test cell. Rinse the material with deionized water. Allow it to hang in an uncontaminated section of the hood to air-dry overnight.

i. Dispose of the UDMH contaminated tubing. Clean the permeation test cell and the hypergol reservoir. Repeat the test for all of the material samples that are specified in 4.7.6.

4.7.7 Trisodium pentacyanoamino ferrate (TPF) method for UDMH.

4.7.7.1 Reagents.

a. Buffer solution, pH 5.4: Boil 1800 mL of distilled water. While boiling, add 15.5 g of anhydrous sodium phosphate diabasic (NS_2HPO_4); stir and dissolve. Add 9.6 g of citric acid ($C_6H_8O_7$); stir and dissolve. Remove from heat; pour into a 2000-mL volumetric flask; rinse the beaker with 100 mL of boiled distilled water. Cool, and make up to 2000 mL with boiled distilled water.

b. Buffer solution for air sampling: Dilute 50 mL of pH 5.4 buffer to 200 mL with distilled water.

c. TPF ($Na_3Fe(CN)_5NH_3$): Dissolve 0.1 g of TPF in 100 mL of distilled water. Prepare fresh daily; recrystallized or freshly prepared stock TPF must be used.

d. Sulfuric acid (H_2SO_4), 12N: Carefully add 336 mL of reagentgrade concentrated H_2SO_4 to 500 mL of distilled water. Cool. Dilute with distilled water to 1 liter.

e. Stock UDMH solution: Using a 100-mL volumetric flask with a rubber serum cap, add 50 mL of pH 5.4 phosphate-citrate buffer to 5 mL of 12N sulfuric acid. Weigh the flask and contents and record the weight. Using a tuberculin syringe, measure approximately 0.12 mL

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of UDMH. Inject this directly into the solution in the inverted volumetric flask. Swirl and reweigh the flask before diluting to 100 mL with additional pH 5.4 buffer. Example:

100-mL flask + 50 mL pH 5.4 buffer + 5 mL H_2SO_4 = 85.8931 g

100-mL flask after 0.12 mL UDMH injected = 85.9931 g

In this case, the resultant solution would contain 1 milligram (mg) of UDMH per milliliter. Due to the volatility of UDMH, it is necessary to follow this procedure to attain an accurate stock standard. This stock-standard UDMH is stable under refrigeration for approximately 2 weeks.

f. UDMH working standard: Take 1 mL of stock UDMH standard and dilute to 100 mL in a volumetric flask with phosphate-citrate buffer, pH 5.4. The resultant solution will now contain approximately 10 micrograms (µg) of UDMH per milliliter (dependent on the stock solution).

4.7.7.2 Calibration curve.

a. For the post-dilution standard concentrations shown below, pipette the following quantities into 25 mL volumetric flasks:

Standard concentration µg per mL	Volume used, mL		
	Working standard	pH 5.4 buffer	TPF
Blank	0	5	5
0.38	1.0	5	5
1.14	3.0	5	5
1.52	4.0	5	5
1.90	5.0	5	5
2.66	7.0	5	5

b. Dilute all samples to 25 mL with distilled water; the resulting solution will have a pH of approximately 5.7.

c. Shake the solutions and let them stand 30 minutes at room temperature.

d. Zero the instrument with a blank, and read the optical density at 550 nm on a Beckman DK-2 spectrophotometer, or equal.

e. Plot a standard calibration curve (optical density versus g per mL of UDMH).

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4.7.7.3 Procedure.

a. Add 12 mL of diluted buffer to a midget impinger.

b. Sample air through the impinger at a flow rate of 1 liter per minute or slightly less.

c. Record the total sample time and flow rate.

d. Add 3 mL of 0.1 percent TPF to the sample and to a blank containing 12 mL of diluted buffer.

e. Zero a Beckman DK-2 spectrophotometer, or equal, at 500 nm with the blank solution.

f. After 30 minutes of reaction time, read the optical density of the sample.

4.7.7.4 Calibrations. Determine the ppm of UDMH in air by means of the following equation:

$$\text{ppm of UDMH} = \frac{(15)\ (G)\ (D)\ 24.45}{(R)\ (T)\ 60}$$

where: G is µg/mL of UDMH from calibration curve

D is the appropriate dilution factor, if needed

R is flow rate (liters per minute)

T is sample time (minute)

* 4.7.8 Alternate method for chemical analysis of fuel.

4.7.8.1 Reagents.

a. Stock UDMH and hydrazine solution; prepare a stock solution containing 7.9 µg per mL of UDMH (MIL-P-25604) and 10.0 µg per mL of N_2H_4 (MIL-P-25604) by standard dilution methods.

b. Derivatizing solution; 2-furaldehyde, 1.0 mL, in 25 mL of sodium acetate solution. (42 g per liter)

c. Ethyl acetate.

d. Sulfuric acid solution, 0.4 N.

e. Naphthalene internal standard; 200 mg dissolved in 100 mL of ethyl acetate (2.0 µg per microliter [µL]).

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f. Sulfuric acid-coated firebrick adsorption tubes.

4.7.8.2 Calibration curve.

a. Pipette 2 mL, 4 mL, and 6 mL samples from the stock solution into 10 mL volumetric flasks. Dilute with 0.4 N sulfuric acid.

b. Aliquots (2 mL) of the three standards are mixed with 2 mL of derivatizing solution.

c. After standing for 1 hour, the derivatized fuels are extracted twice with 1 mL portions of ethyl acetate.

d. Five µL (2.0 µg per µL) of naphthalene internal standard are added to 1 mL of the ethyl acetate extract.

e. An aliquot, 2 µL, of this solution is then injected into a Hewlett Packard 5992 gas chromatograph/mass spectrometer (GC/MS). The derivatized fuels are analyzed by programming the GC/MS with Selected Ion Monitoring (SIM), which monitors the abundance of ions of specific masses as selected by the user. The gas chromatography separations are accomplished using a 2 foot long column with a 2 mm inside diameter glass column packed with 60-80 mesh Tenax. The GC/MS conditions are given in table VI. The SIM program normalizes the abundance of the masses of the selected fuel derivative ions by converting them to percentages of the abundance of the mass of the selected naphthalene ion.

TABLE VI. GC/MS instrument conditions for N_2H_4 and UDMH analyses.

Injection port temperature	250°C
Temperature program start temperature	200°C
Temperature program final temperature	280°C
Time at start temperature	0.2 minute
Temperature programming rate	15°C per minute
Time at final temperature	2.0 minutes
Solvent elution time	1.9 minutes
Ion window size	0.05

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TABLE VI. GC/MS instrument conditions for N_2H_4 and UDMH analyses (Continued).

SIM data handling parameter	
Integrate sensitivity	0.05
Area threshold	20
Smoothing factor	1.0
Oversize factor	3.0

f. The calibration curve is prepared by plotting the normalized ion abundance as a function of the concentrations of UDMH and N_2H_4. (See figures 7 and 8).

4.7.8.3 Procedure.

a. After each fuel permeation test is completed (4.7.8), empty the contents of the firebrick tube into 3 mL of water and allow to stand for 2 hours to desorb the fuels.

b. Aliquots (2 mL) of the solutions are then derivatized, extracted, and analyzed by GC/MS following steps 4.7.9.2b through 4.7.9.2e.

4.7.8.4 Calculations. Determine the fuel permeation rate mg per inch² per hour by means of the following equation:

$$R = \frac{(Res)\ (V)}{(C)\ (S)\ (A)}$$

where: R is the fuel permeation rate

Res is GC/MS response (normalized abundances)

V is the volume of water into which tube contents were absorbed, 3 mL

C is conversion to mg, 1000 µg per mg

S is the slope of the appropriate calibration curve

A is the area of fabric exposed to A-50 3.14 inch²

5. PACKAGING

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.3).

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5.1.1 Levels A, B and C. The cloth shall be put up in rolls and packaged in accordance with the applicable requirements for class 2 fabrics of PPP-P-1136.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.3).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136, except that only fiberboard or rigid type shipping containers shall be used. In addition, an A-flute double-faced corrugated fiberboard separator pad having a minimum bursting strength of 200 psi conforming to type I, class 1 of PPP-B-636 and measuring one-fourth of an inch less than the inside length and width dimensions of the shipping container shall be placed between each layer of rolls.

5.3 Marking. In addition to any special marking required by the contractor or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

5.3.1 Precautionary marking. The following precautionary marking, in letters not less than three-fourths of an inch in height, shall appear at least on one side and wherever practicable on two sides of each shipping container.

KEEP COOL AND DRY.
DO NOT STAND ON END.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended * for use in the fabrication of missile propellant handler's protective clothing.

6.2 Source of material. A material meeting the requirements of this specification is identified as style YX1002 manufactured by DuPont Company, F&F Department, Industrial Coated Fabric Section, Wilmington, Delaware 19898.

6.3 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of the specification.

b. Width required (see 3.3).

c. Length required (see 3.4).

d. Levels of packaging and packing (see 5.1 and 5.2).

6.4 Charring point test apparatus. An apparatus of the type described in 4.6.2.1 is manufactured by the Nalge Company, Rochester, New York.

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* 6.5 Liquid or gas collecting medium. The hazardous liquid permeation test cell described in figures 5 and 7 is manufactured by Pesce Lab Sales, Box 235, Kennett Square, Pennsylvania 19348.

6.6 Change from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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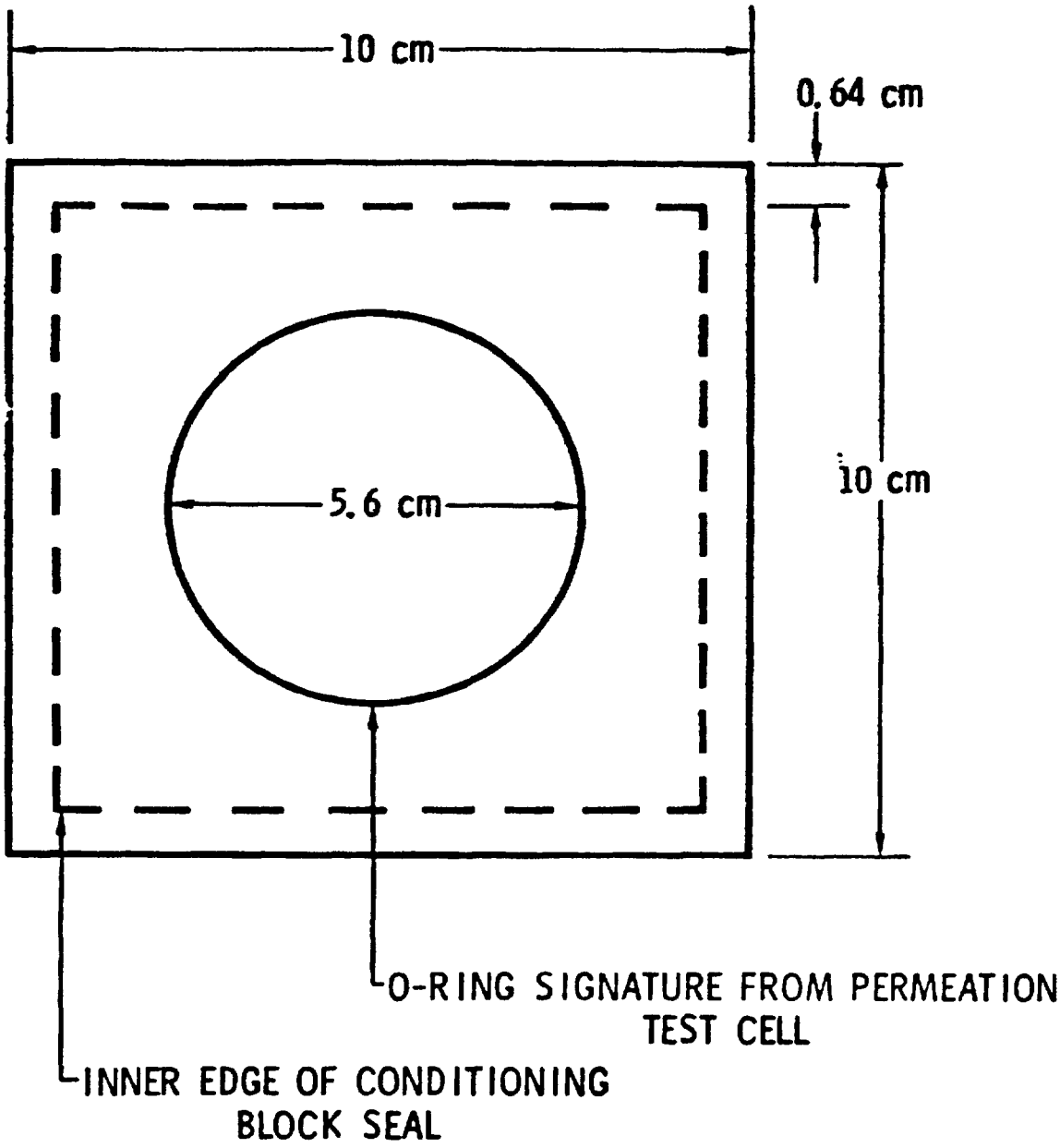


FIGURE 1. Test material.

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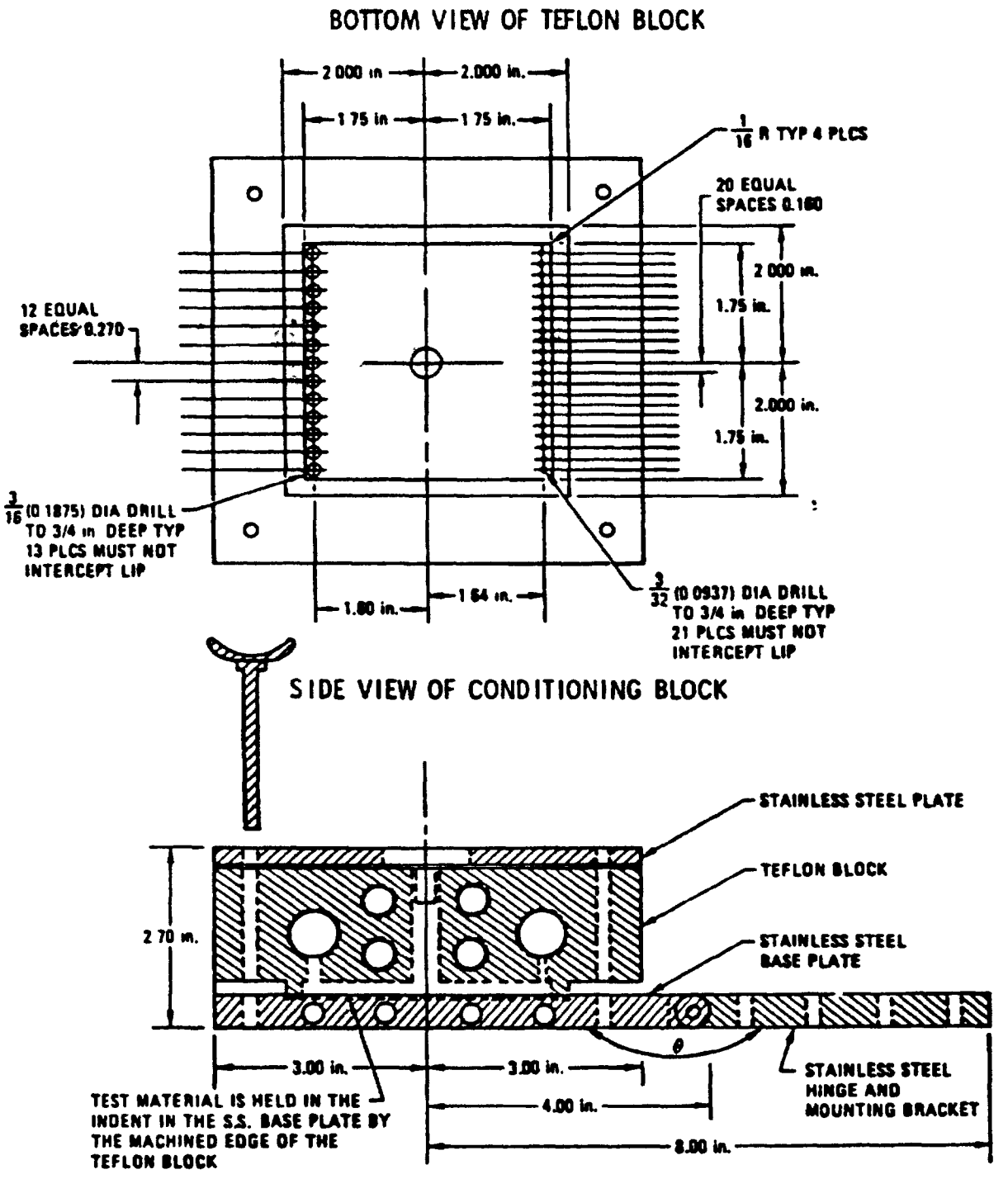


FIGURE 2. Detailed drawing of the conditioning block.

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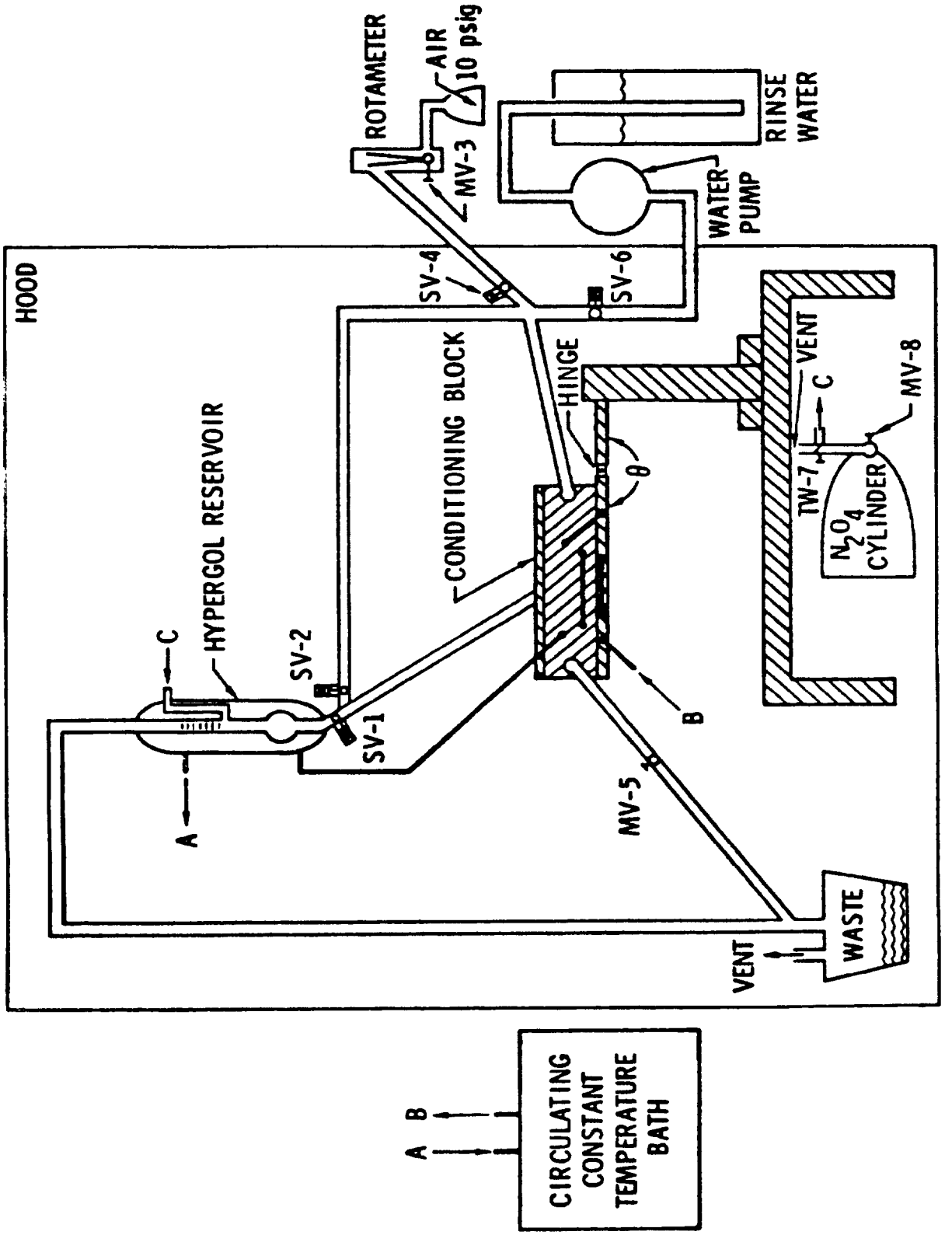


FIGURE 3. The N_2O_4 conditioning apparatus solenoid valve (SV), manual valve (MV), and three-way valve (TW).

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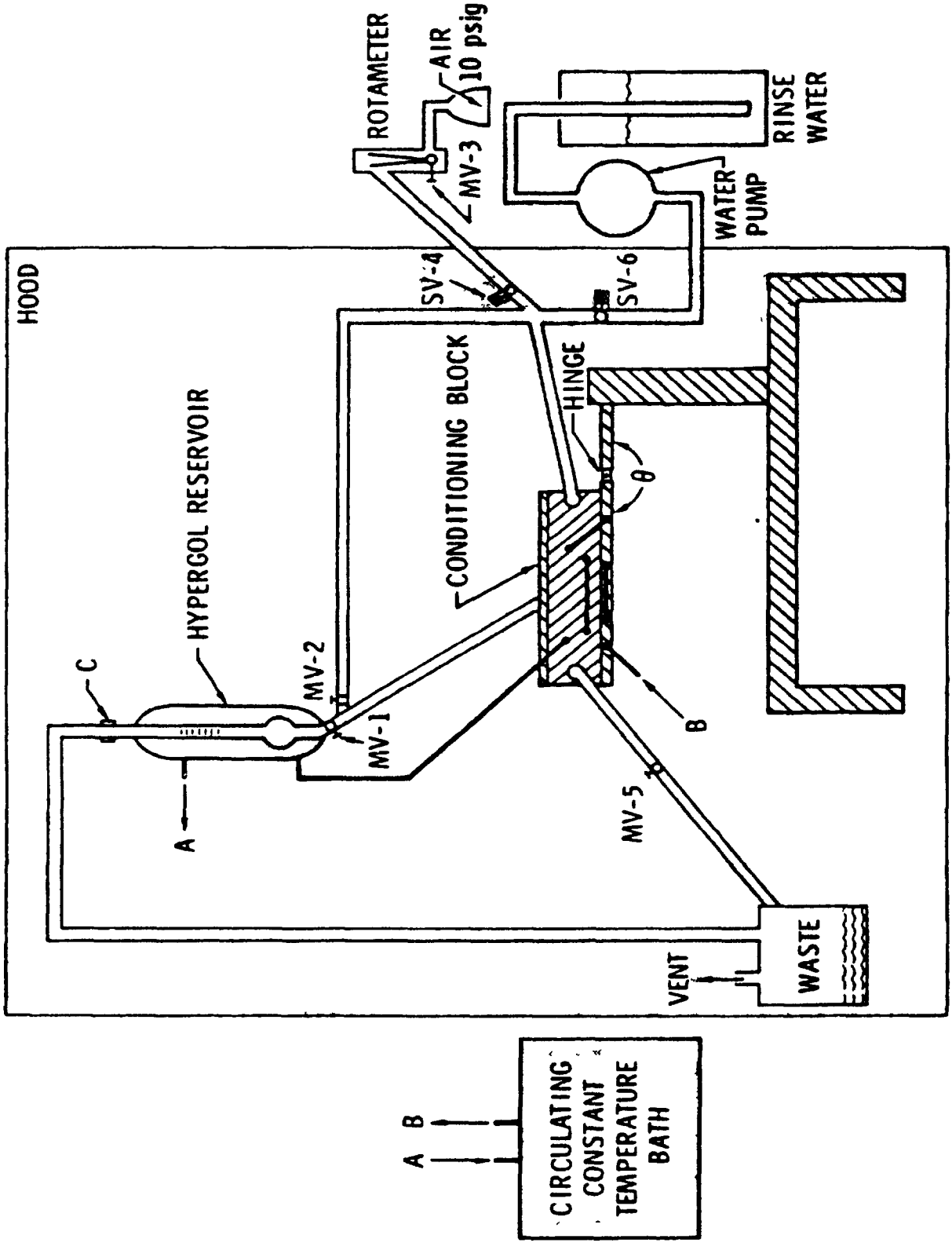


FIGURE 4. The fuel conditioning apparatus solenoid valve (SV) and manual valve (MV).

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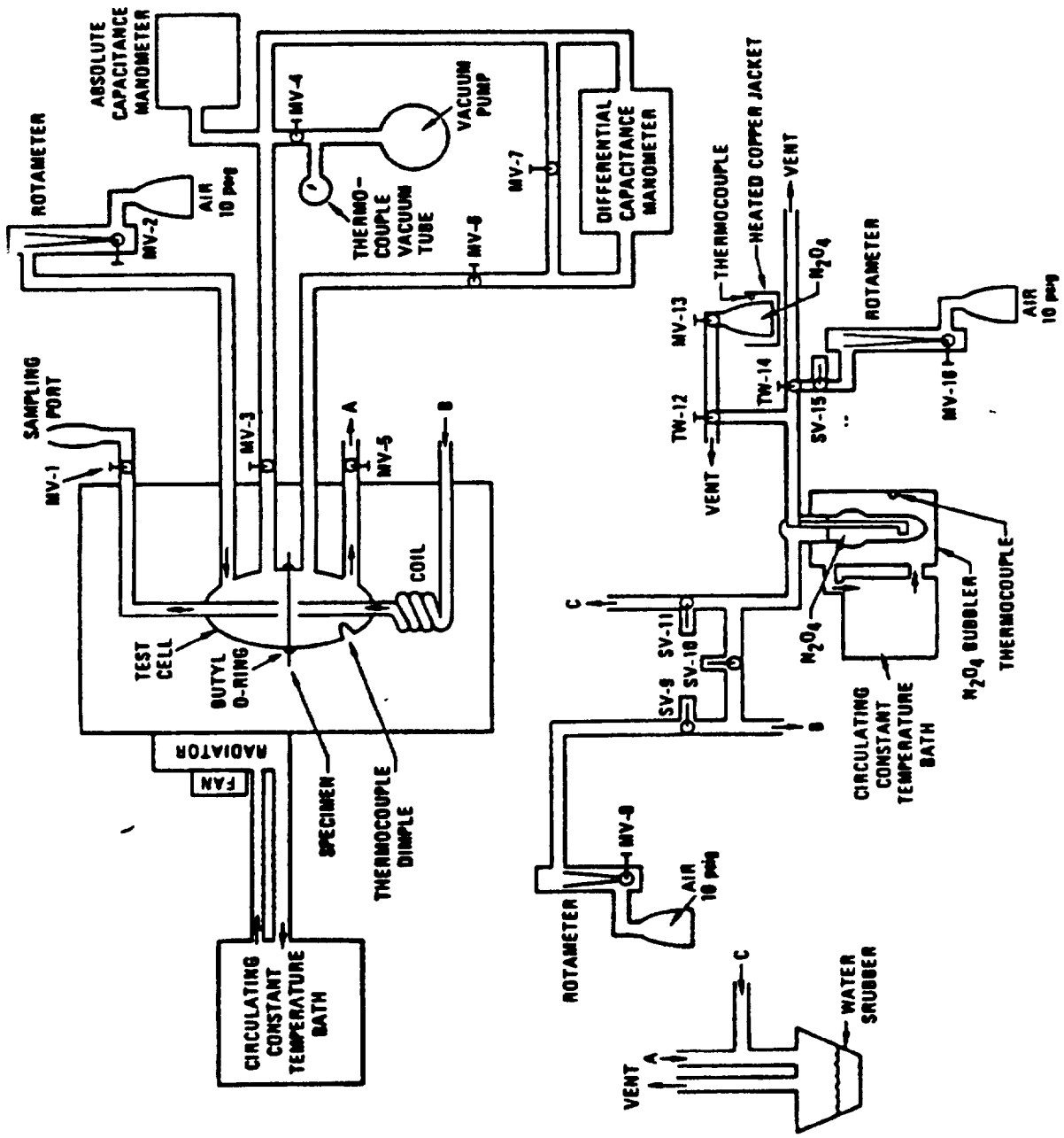


FIGURE 5. The N_2O_4 permeation test apparatus solenoid valve (SV), manual valve (MV), and three-way valve (TW).

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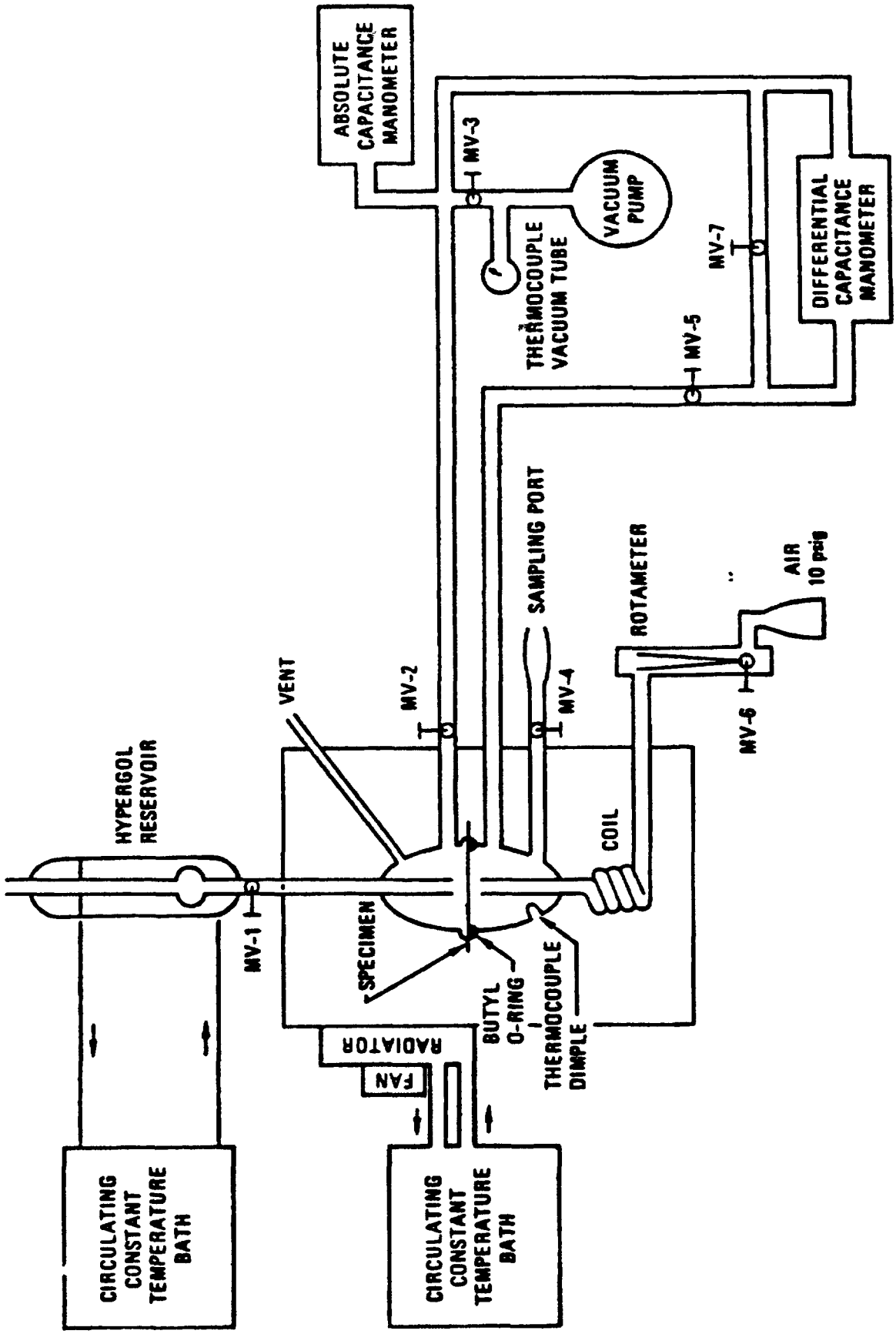


FIGURE 6. The fuel permeation test apparatus manual valve (MV).

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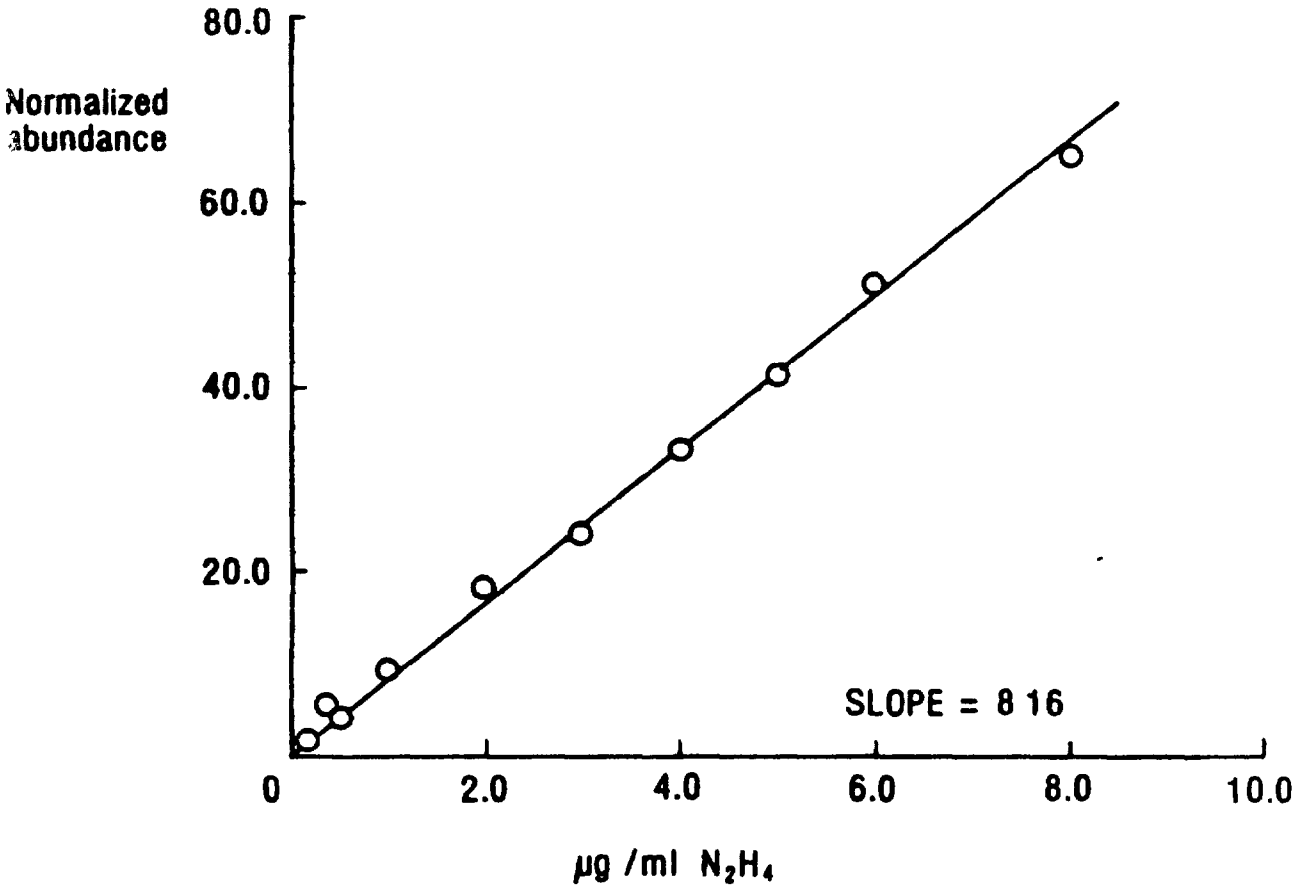


FIGURE 7. Calibration curve for N_2H_4 determination.

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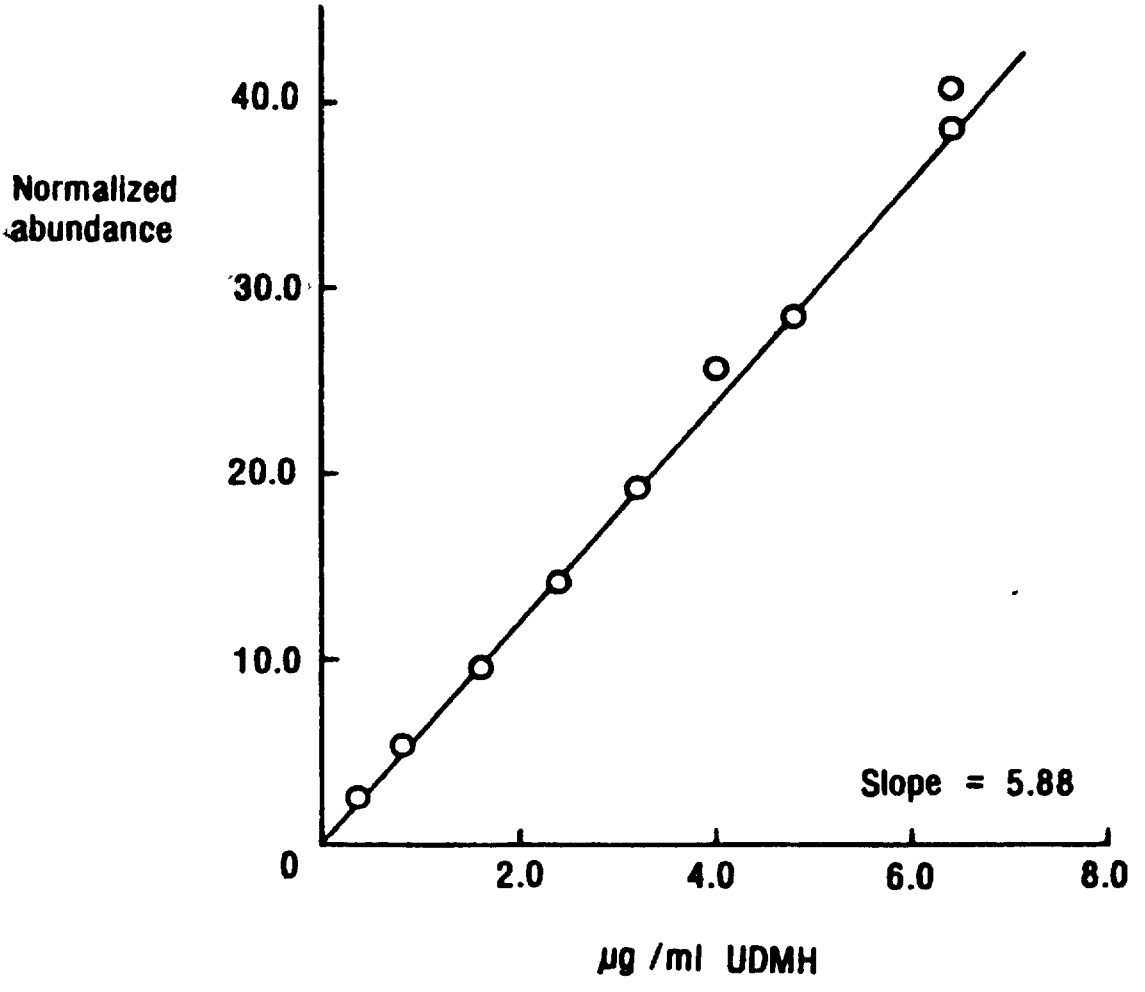


FIGURE 8. Calibration curve for UDMH determination.

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	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

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